

ROUND 2

Technical Leadership Evaluation

Problem Statements 3 & 4

EVALUATION SCOPE	2 Problem Statements 6 Questions Round 2 of 3
TARGET EXPERIENCE	3 years shipping AI/ML systems in production
ASSESSMENT TIME	45–60 minutes
FRAMEWORK VERSION	v2.1 Professional Edition

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ROUND OVERVIEW

This is **Round 2 of a 3-round evaluation process**. You will be assessed on 2 problem statements covering 3 distinct technical areas, with 6 questions total. **What to expect:** • 2 problem statements (context + core challenge) • 6 targeted questions (3 per problem statement) • Escalating difficulty: Architecture → Production → Cost • 45–60 minutes to complete **Format:** Each problem statement contains 3 questions: – Question A (12–15 min): Architecture & Design – Question B (10 min): Production Challenges & Operations – Question C (8–10 min): Cost/Scale Optimization **Scoring:** Each question scored 1–5 per detailed rubric. Your section score is the average of all 6 questions in this round. **After Round 3:** You'll be invited to submit an engineering demo (10–15 min video) and participate in a Founder Round crisis scenario. These final components are worth 40% of your overall evaluation score.

PROBLEM STATEMENT 3

Person Recognition & Identity Intelligence

Context:

In a distributed surveillance system, the same person may appear across multiple cameras over time. System should link appearances, building an identity graph.

Core Problem:

Build a scalable, performant person identification system within privacy, governance, and operational constraints.

Question 3.1: Scale, Performance & Architecture

Time: 12 min | Character Limit: 4,500 characters

Design a person identification system scaling from 1,000 to 1 million identities. Include: a) Would you use face recognition, person re-identification (ReID), or both? Why? b) How would your database architecture change as identities scale? (1K → 10K → 100K → 1M) c) Estimate query latency at each scale. At what scale does vector DB become necessary? d) How would you update identity embeddings as people age or change appearance? e) What would be the storage cost per identity at 1 million scale?

SCORING RUBRIC (1–5 scale)

5 Points – Exceptional: Justifies recognition approach with trade-offs. Discusses embedding vs. dense storage. Understands vector DB fundamentals and when needed. Addresses representation drift. Provides storage estimates with reasoning. **4 Points – Strong:** Good architecture. Discusses 2–3 scale transitions. Understands core challenges. **3 Points – Adequate:** Basic approach. Vague on scale transitions. **2 Points – Weak:** Incomplete architecture. **1 Point – Poor:** Fundamental misunderstanding.

Evaluator Notes

- Do they understand face recognition and ReID solve different problems?
- Have they thought about representation drift?
- Do they know what vector DBs are actually for?
- Do they understand scaling challenges of similarity search?

Question 3.2: Privacy, Governance & Compliance

Time: 10 min | Character Limit: 3,500 characters

Your person identification system will process millions of face/identity captures. Design your privacy and governance framework: a) Consent management: Who can be tracked, and how do you enforce consent at scale? b) Audit trail: What would you log, and how would you query audit logs? c) Data retention: How would you implement time-based deletion at scale? d) What are the hardest technical problems in privacy-compliant person tracking?

SCORING RUBRIC (1–5 scale)

5 Points – Exceptional: Discusses consent mechanisms at tech level (database design, APIs). Addresses audit trail architecture. Understands retention enforcement challenges. Identifies hard problems (false matches,

re-identification risk). **4 Points – Strong:** Good governance framework. Addresses 3+ aspects. **3 Points – Adequate:** Basic privacy thinking. Covers 2 aspects. **2 Points – Weak:** Incomplete framework. **1 Point – Poor:** Doesn't take privacy seriously.

Evaluator Notes

- Do they treat privacy as architecture, not compliance afterthought? • Have they dealt with GDPR/DPDP-like regulations?
- Do they understand audit trail complexity at scale? • Do they acknowledge hard problems (false matches = privacy violations)?

Question 3.3: False Positive Tolerance & Remediation

Time: 8 min | Character Limit: 3,000 characters

A customer reports: "System matched me (John) with David. We look similar, but different people. Your system flagged me 5 times as David last week." This false positive eroded trust. What is your: a) Root cause? (Model? Database? Matching threshold?) b) Immediate fix to prevent this specific match? c) Systemic change to reduce similar false positives?

SCORING RUBRIC (1–5 scale)

5 Points – Exceptional: Distinguishes detection FP vs. matching FP. Proposes immediate technical fix (manual exclusion, threshold adjustment). Proposes systemic change (hard example retraining, similarity thresholding, manual review). **4 Points – Strong:** Good root cause analysis. Proposes immediate and systemic fixes. **3 Points – Adequate:** Identifies problem. Suggests one fix. **2 Points – Weak:** Surface-level answer. **1 Point – Poor:** Misses core issue.

Evaluator Notes

- Do they understand difference between FP (wrong detection) and FN (missed match)? • Have they dealt with high-stakes matching problems? • Do they understand privacy/trust implications? • Do they propose measurable solutions?

PROBLEM STATEMENT 4

Object & Asset Recognition

Context:

System must identify and track objects: equipment, vehicles, packages, safety gear, or operational assets. Customers care about different object classes.

Core Problem:

Build a scalable, customizable object detection system that handles customer-specific classes and rapid introduction of new classes.

Question 4.1: Model Strategy & Class Scaling

Time: 12 min | Character Limit: 4,500 characters

You need to detect operational assets: hard hats, safety vests, forklifts, packages, vehicles, uniforms, etc. Start with 20 classes. Within a year, you may have 200+ classes across different customer verticals. Design your detection strategy: a) Single monolithic YOLO/Faster R-CNN vs. modular approach? b) At what point does embedding-based retrieval become better than classification? c) How would you introduce a new customer-specific class in production? (Process, data requirements, timeline) d) Compare: – Strategy A: Single large model, 500 classes, 87% mAP, 200ms inference – Strategy B: Tiered approach (coarse → fine), 500 classes, 92% mAP, 350ms inference Which would you choose? Justify.

SCORING RUBRIC (1–5 scale)

5 Points – Exceptional: Discusses monolithic vs. modular with trade-offs. Understands when embeddings outperform classification. Has concrete process for new classes (data collection → validation → retraining). Compares strategies quantitatively. **4 Points – Strong:** Good model strategy. Discusses class scaling thoughtfully. Some process for new classes. **3 Points – Adequate:** Basic approach. Vague on scaling. **2 Points – Weak:** Incomplete strategy. **1 Point – Poor:** Fundamental confusion.

Evaluator Notes

• Do they understand difference between classification (closed set) and embedding-based search (open set)? • Have they deployed multi-class detection at scale? • Do they think about inference cost as they scale classes? • Do they understand when a single model breaks?

Question 4.2: Continuous Learning & Model Updates

Time: 10 min | Character Limit: 3,500 characters

You deploy new object detection model (87% → 92% accuracy) to 1,000 customers. Two weeks later, three customers report issues: – Customer A: 2% more false positives (harder to tolerate) – Customer B: System is slower (350ms vs. 200ms) – Customer C: Loves it, no issues Walk through: a) Process for catching this deployment issue earlier b) How you would roll back for specific customers while rolling forward for others c) Your model versioning and A/B testing strategy d) How you would have validated before general deployment

SCORING RUBRIC (1–5 scale)

5 Points – Exceptional: Proposes canary deployment. A/B testing on subset. Clear rollback procedure. Per-customer feature flags. Has validation plan before rollout. **4 Points – Strong:** Good deployment strategy. Addresses rollback and A/B testing. **3 Points – Adequate:** Basic deployment thinking. Addresses 2 aspects. **2 Points – Weak:** Incomplete strategy. **1 Point – Poor:** No deployment awareness.

Evaluator Notes

- Do they think about safe rollout practices?
- Have they operated feature flags for ML models?
- Do they understand per-customer rollback?
- Do they know how to measure A/B test results?

Question 4.3: Cost Optimization & Efficiency

Time: 8 min | Character Limit: 3,000 characters

Inference cost is your dominant variable cost. Reduce it by 50% while maintaining 90%+ accuracy. Discuss:
a) Model compression techniques: quantization, pruning, distillation
b) Multi-stage inference: coarse → fine (skip fine detection for unlikely cases)
c) Caching: would you cache detections? When is this safe?
d) What's your trade-off analysis?

SCORING RUBRIC (1–5 scale)

5 Points – Exceptional: Discusses multiple techniques with quantitative estimates (e.g., "int8 quantization gives 4x speedup with <2% accuracy loss"). Understands latency-accuracy-cost triangle. Proposes multi-pronged approach. **4 Points – Strong:** Good discussion of 2–3 techniques with some quantitative reasoning. **3 Points – Adequate:** Mentions techniques but vague on impact. **2 Points – Weak:** Limited optimization ideas. **1 Point – Poor:** No optimization thinking.

Evaluator Notes

- Have they actually implemented model compression?
- Do they understand real-world speedups vs. theoretical?
- Do they think about where optimization has most impact?
- Do they propose measurable solutions?

SCORING FOR THIS ROUND

This Round's Score: Round 2 Score = Average of Questions 3.1, 3.2, 3.3, 4.1, 4.2, 4.3 Each question is scored 1–5 per the rubric above. Your section score is the simple average. **Overall Evaluation (After All 3 Rounds + Demo + Founder Round):** Overall Score = (Section Average $\times$ 0.60) + (Demo $\times$ 0.15) + (Founder Round $\times$ 0.25) Where: • Section Average = Average of Rounds 1, 2, and 3 • Demo = Separate 1–5 score (10–15 min video) • Founder Round = Separate 1–5 score (crisis scenario, 25% weight) **Key Insight:** The Founder Round is the most important component (25% weight). It separates operators who've debugged production incidents from architects who've only designed systems theoretically.